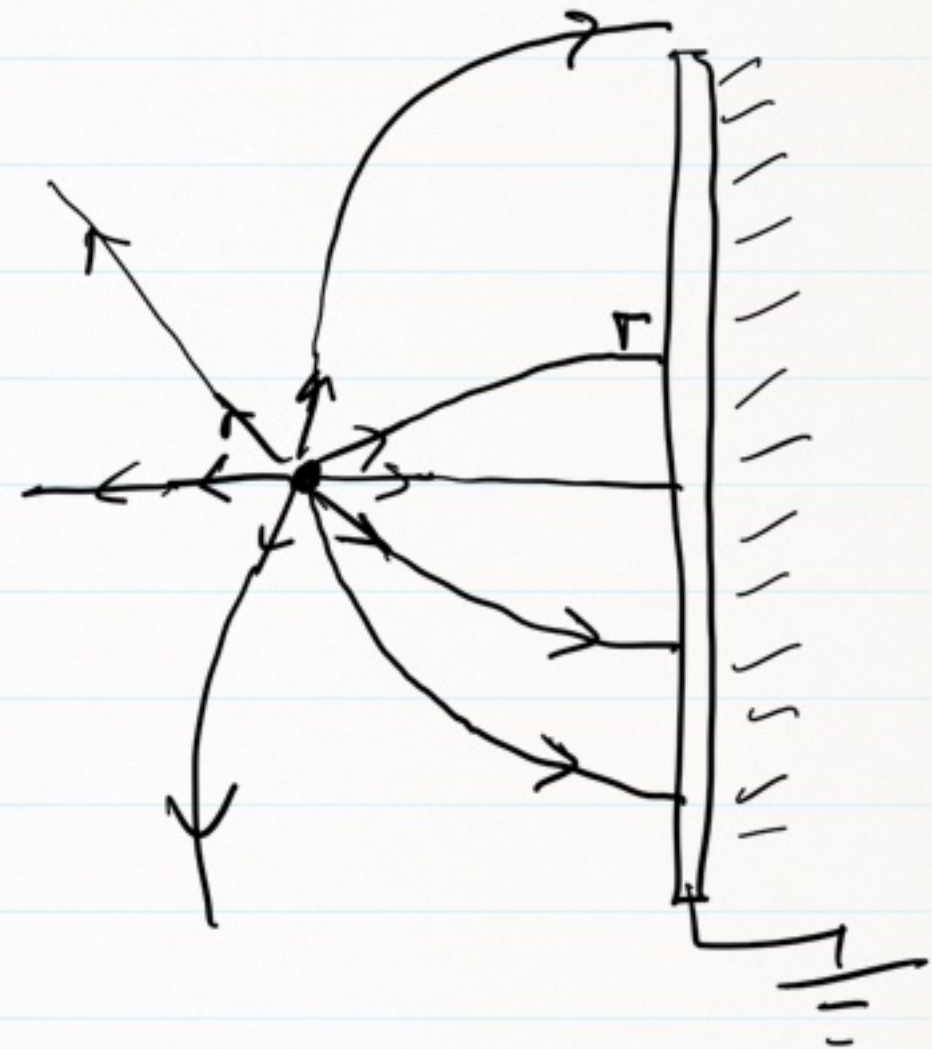
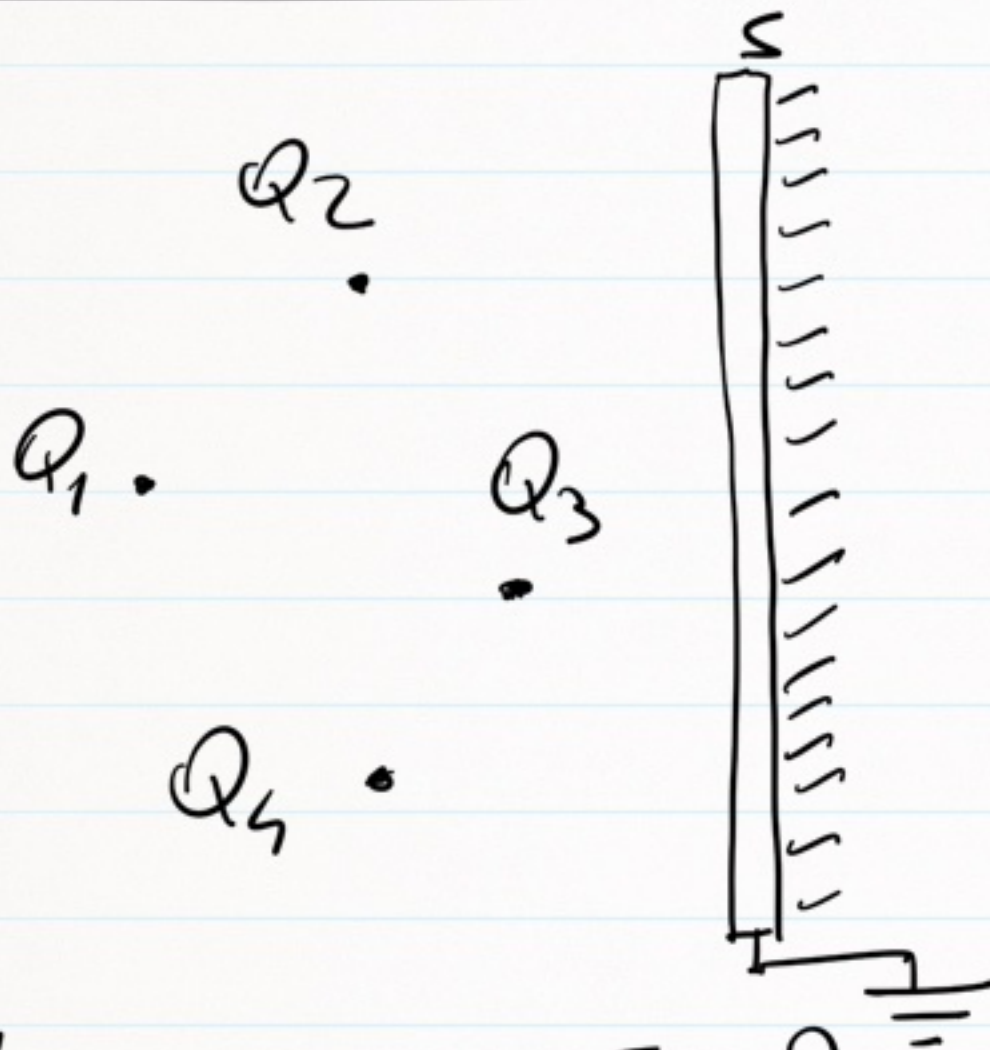


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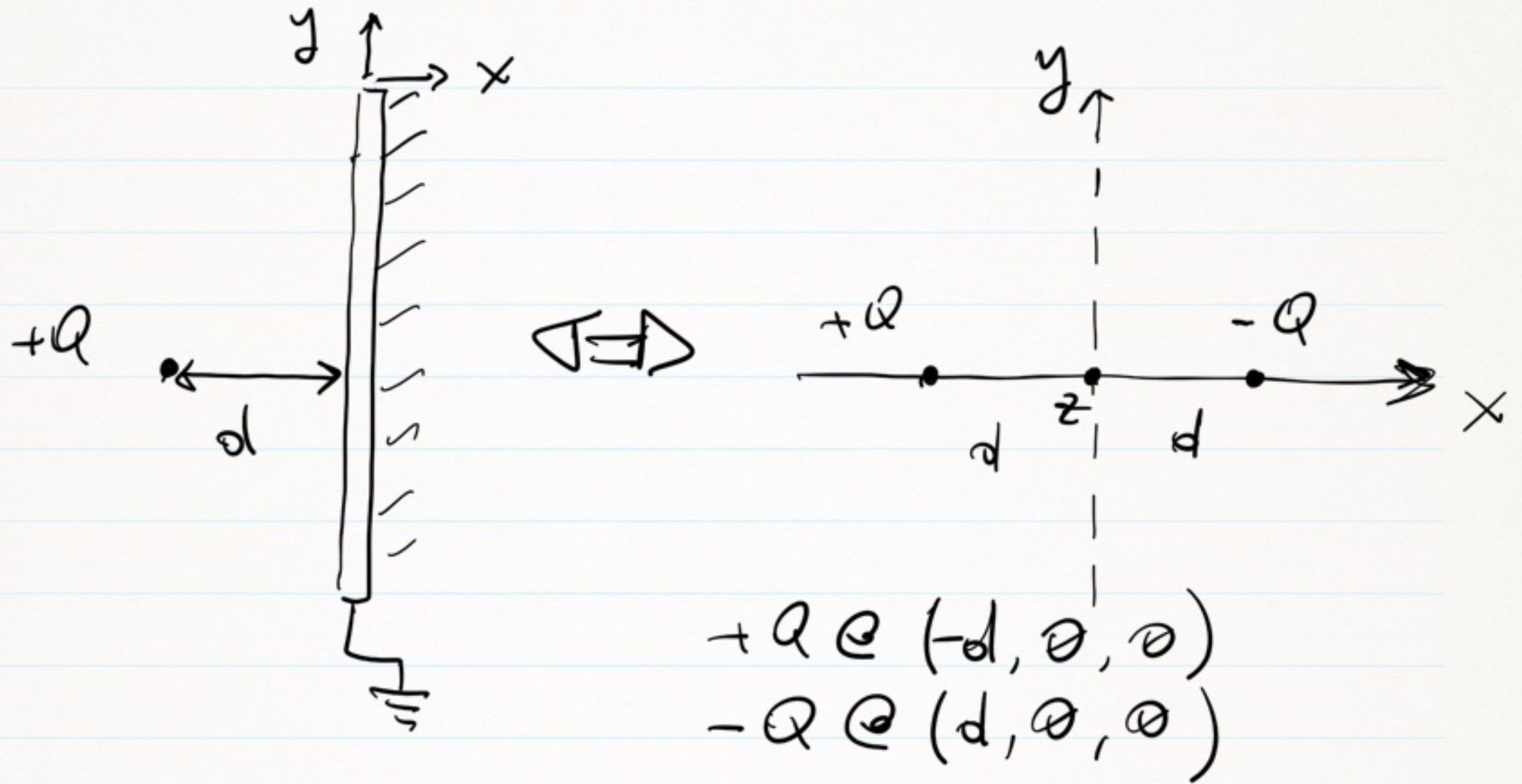
METODO CARICHE IMMAGINI



$\rho_{indotta}$ su S ?

$$\nabla^2 V = - \frac{\rho}{\epsilon_0} \quad \text{!}$$

$$\left\{ \begin{array}{l} Q_i @ (x_i, y_i, z_i) \\ V_S = \text{cost. } (= 0) \\ \lim_{r \rightarrow \infty} V(r) = 0 \end{array} \right.$$



$$\Rightarrow V_{y_2^A} = 0$$

① $\rho(x, y, z)$

$$\bullet V_p = \frac{Q}{4\pi\epsilon_0} \left\{ \frac{1}{\sqrt{(x+d)^2 + y^2 + z^2}} - \frac{1}{\sqrt{(x-d)^2 + y^2 + z^2}} \right\}$$

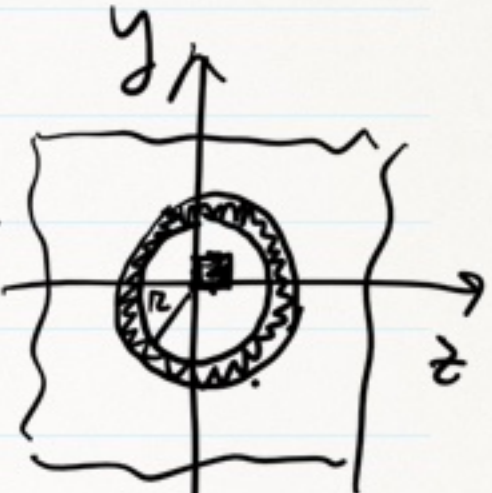
$$E_x = - \frac{\partial V}{\partial x} = - \frac{Q}{4\pi\epsilon_0} \left\{ \left(-\frac{1}{2}\right) \frac{2(x+d)}{[(x+d)^2 + y^2 + z^2]^{3/2}} - \left(-\frac{1}{2}\right) \frac{2(x-d)}{[(x-d)^2 + y^2 + z^2]^{3/2}} \right\}$$

$\bullet E_x(0, y, z) \xrightarrow{\text{at } x=0} \frac{Q}{2\pi\epsilon_0} \cdot \frac{d}{(d^2 + y^2 + z^2)^{3/2}}$

$$\bullet E_x = - \frac{\sigma}{\epsilon_0} \Rightarrow \sigma(y, z) = - \epsilon_0 E_x$$

$$\sigma(y, z) = - \frac{Q}{2\pi} \cdot \frac{d}{(d^2 + y^2 + z^2)^{3/2}}$$

max @ $y=z=0$

$$\int \rho \, dy \, dz = - \frac{Qd}{2\pi} \int \frac{dy \, dz}{(d^2 + y^2 + z^2)^{3/2}} =$$


$$= - \frac{Qd}{2\pi} \int_0^{\infty} \frac{2\pi r \, dz}{(d^2 + r^2)^{3/2}} = - \frac{Qd}{2} \left[- \frac{2}{(d^2 + r^2)^{1/2}} \right]_0^{\infty} =$$

$$= - \frac{Qd}{2} \left[0 + \frac{2}{d} \right] = -Q$$

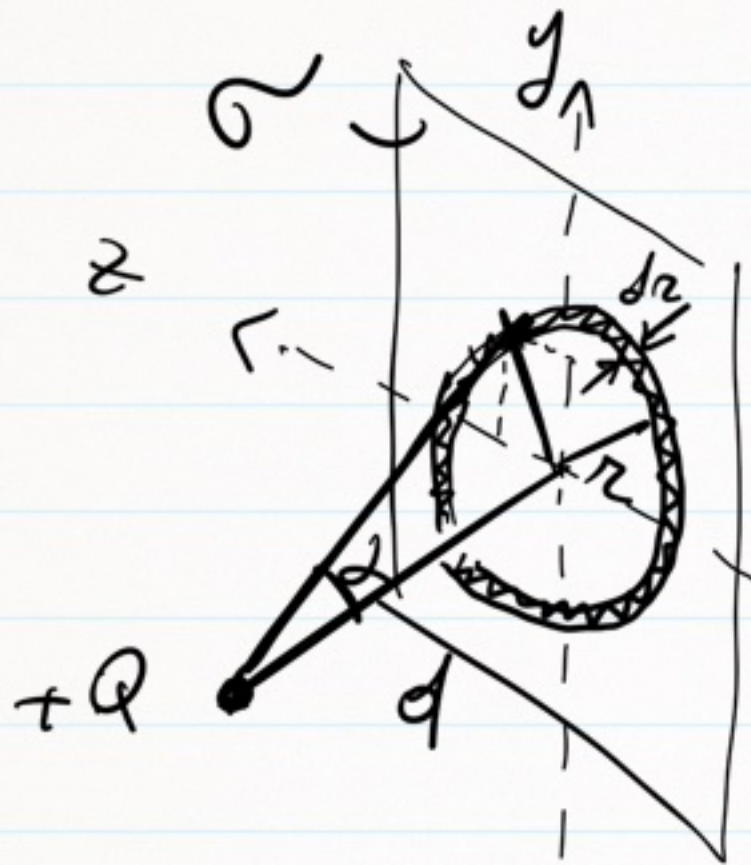
Campo generado @ $x = +d$ where $-Q$:

$$E_x = - \frac{Q}{4\pi\epsilon_0} \cdot \frac{(x-d)}{[(x-d)^2 + y^2 + z^2]^{3/2}}$$

$$\text{@ } x = -d \quad \frac{Q}{4\pi\epsilon_0} \cdot \frac{2d}{(-2d)^{3/2}}$$

($y=z=0$)

$$F_x = Q \cdot E_x = \frac{Q^2}{4\pi\epsilon_0 \cdot 4d^2} = \frac{Q^2}{16\pi\epsilon_0 d^2}$$



$$dE = \frac{1}{4\pi\epsilon_0} \frac{\sigma dS}{(d^2 + r^2)^{3/2}}$$

$$= -\frac{1}{4\pi\epsilon_0} \cdot \frac{Q}{2\pi} \cdot \frac{d \, dx \, dy}{(d^2 + y^2 + z^2)^{3/2}}$$

$$= \frac{1}{(d^2 + y^2 + z^2)} \cdot \cos\alpha$$

$$\cos\alpha = \frac{d}{\sqrt{d^2 + y^2 + z^2}}$$

$$= -\frac{1}{4\pi\epsilon_0} \frac{Q}{2} d^2 \frac{2r \, dr}{(d^2 + r^2)^3}$$

$$\int_0^\infty dE = -\frac{1}{4\pi\epsilon_0} \frac{Q}{2} d^2 \left[-\frac{1}{2} \frac{1}{(d^2 + r^2)^2} \right]_0^\infty = \frac{Q}{16\pi\epsilon_0 d^2}$$

$\frac{1}{2d^2}$

$$n = \# \text{ coriche puntiformi} / m^3$$

(q)

$$\vec{p}_i = q_i \Delta \vec{x} \quad , \quad \vec{p}_0 = n (q \Delta \vec{x})$$

$$\underline{p_0 = \rho \Delta x} \quad \Rightarrow \quad \rho = p_0 / \Delta x$$

• $r \geq R$

$$|\vec{p}| = |\Delta \vec{x}| \cdot Q = \Delta x \frac{4\pi R^3}{3} \rho = \frac{4\pi R^3}{3} p_0$$

$$V = \frac{1}{4\pi\epsilon_0} \frac{\vec{p} \cdot \vec{r}}{r^3} = \frac{R^3}{3\epsilon_0} \frac{p_0 \cos\theta}{r^2} \stackrel{r \cos\theta = x}{=} \frac{R^3 p_0 x}{3\epsilon_0 r^3}$$

$$V(r=R) = p_0 x / 3\epsilon_0$$

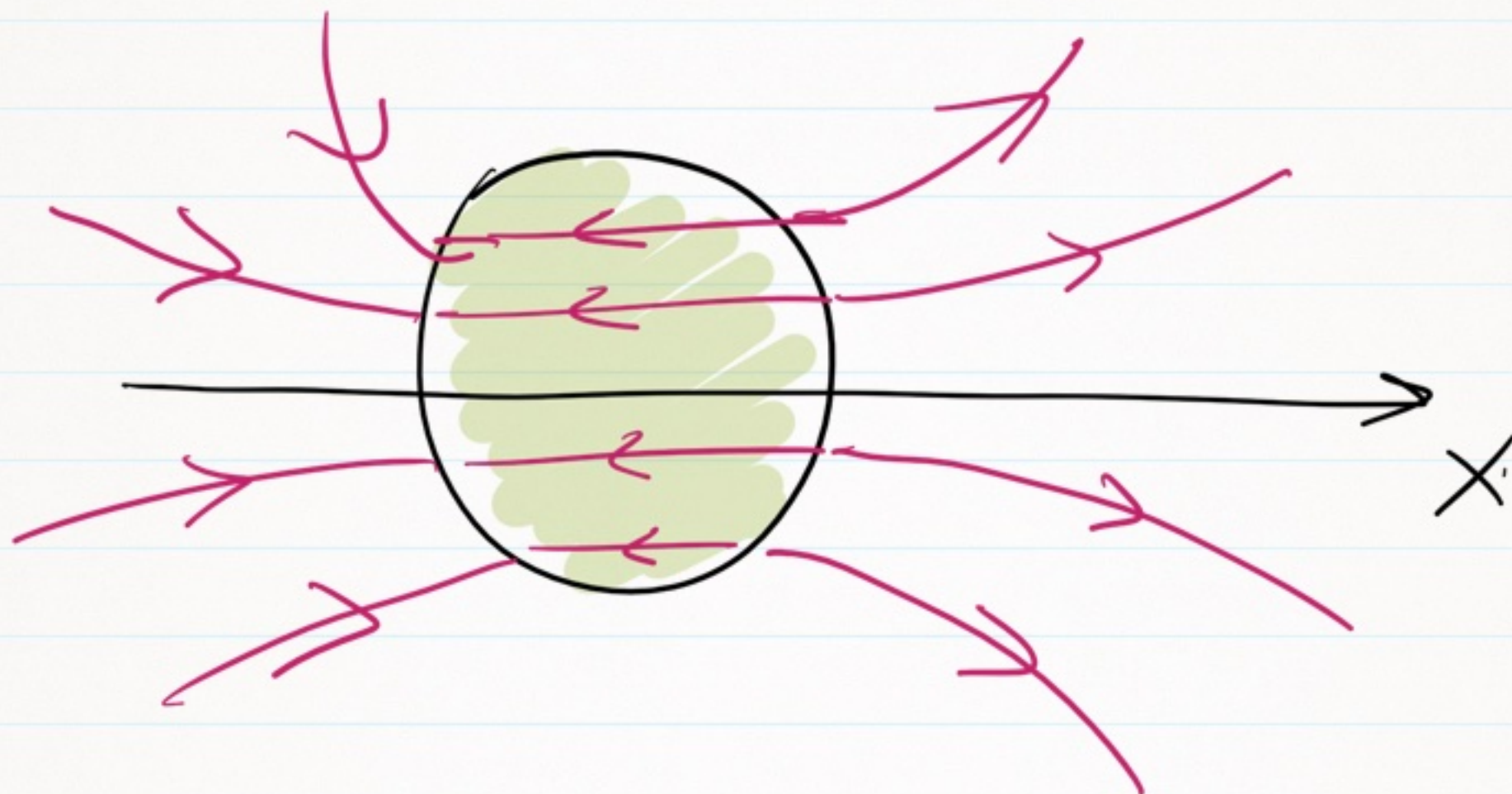
• $r < R$

$$\stackrel{(1)}{\rightarrow} \nabla V = 0 \Rightarrow V(r < R) = V(r = R)$$

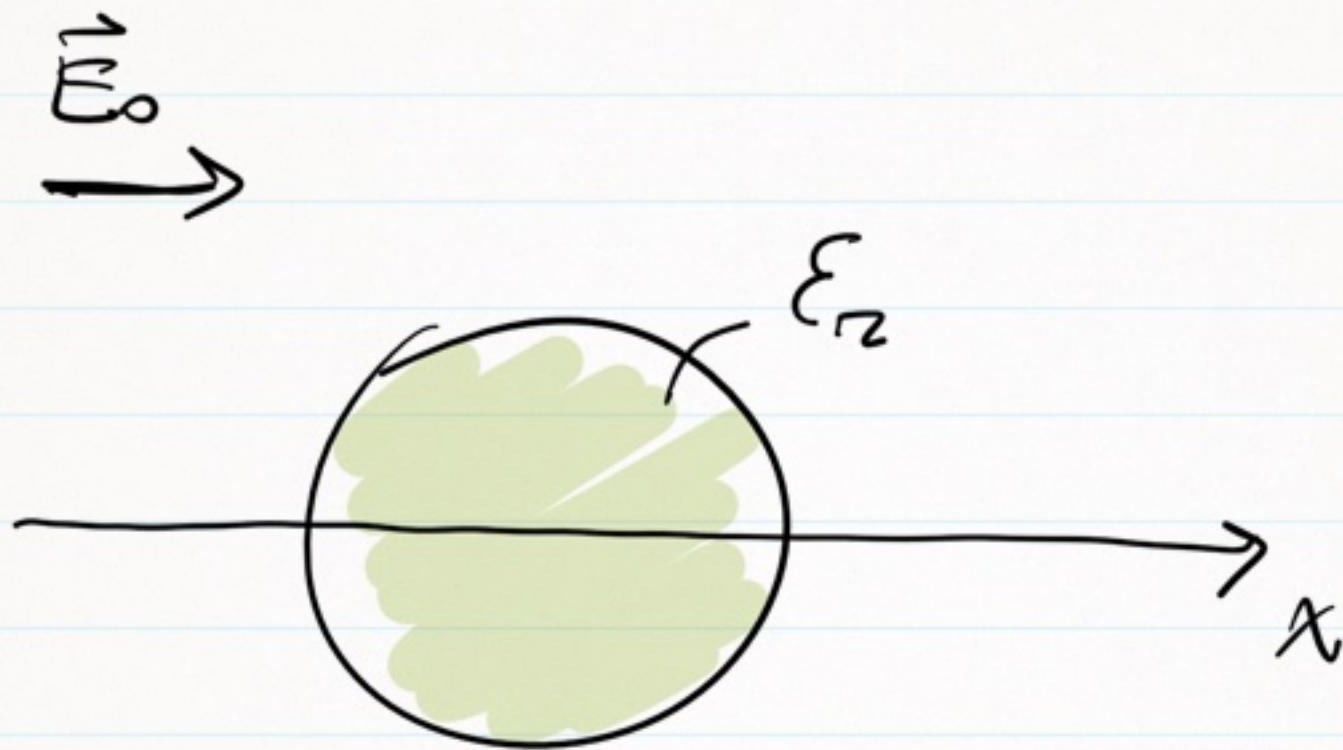
$$\vec{E} = -\vec{\nabla} V \Rightarrow \vec{E} = -\frac{\vec{p}_0}{3\epsilon_0}$$

$$\stackrel{(2)}{\rightarrow} \vec{E}_{-e} = -\frac{e \vec{r}_-}{3\epsilon_0} \quad \vec{E}_{+e} = +\frac{e \vec{r}_+}{3\epsilon_0}$$

$$\begin{aligned} \vec{E} &= \vec{E}_{-e} + \vec{E}_{+e} = \frac{e}{3\epsilon_0} (\vec{r}_+ - \vec{r}_-) = \\ &= -\frac{e \Delta \vec{x}}{3\epsilon_0} = -\frac{\vec{p}_0}{3\epsilon_0} \end{aligned}$$



③



$\vec{E}_0 \rightarrow \vec{P}_0$ (uniforme nel dielettrico, per
assunzione)

$$\vec{P}_0 \rightarrow \vec{E}_{pol} = - \frac{\vec{P}_0}{3\epsilon_0}$$

$z < R$: $\vec{E}_{diel} = \vec{E}_0 + \vec{E}_{pol} = \vec{E}_0 - \frac{\vec{P}_0}{3\epsilon_0}$

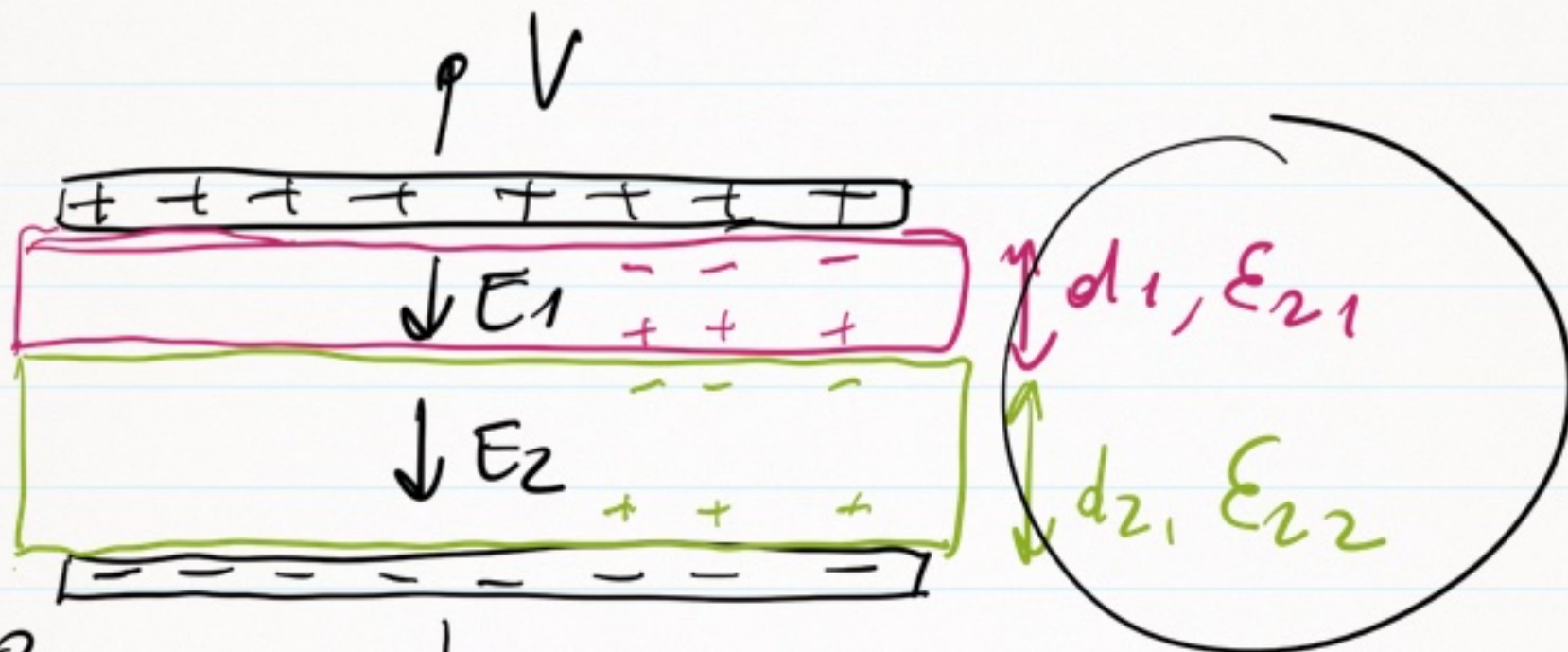
$\vec{P}_0 = \epsilon_0 (\epsilon_2 - 1) \vec{E}_{diel}$ ↗

↘ $\vec{E}_{diel} = \frac{3}{\epsilon_2 + 2} \vec{E}_0$ ($\epsilon_2 > 1$, $E_{diel} < E_0$)

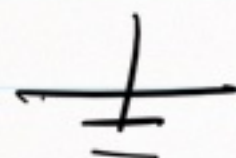
$$\underline{r > R} : |\vec{p}| = \frac{4\pi R^3}{3} \rho_0$$

$$\vec{E}_{out} = \vec{E}_0 + \frac{1}{4\pi\epsilon_0 r^3} \left[3(\vec{p} \cdot \vec{u}_r) \vec{u}_r - \vec{p} \right]$$

④



$$C = \frac{Q}{V}$$



(a) $\vec{D} \perp \text{surface} \Rightarrow D$ $D_1 = D_2 = D$

$\left(E = \frac{\sigma}{\epsilon} \Rightarrow D = \sigma \right)$

$$D = \frac{Q}{S}$$

$$E_1 = \frac{D_1}{\epsilon_0 \epsilon_{21}} = \frac{Q}{\epsilon_0 \epsilon_{21} S}, \quad E_2 = \frac{D_2}{\epsilon_0 \epsilon_{22}} = \frac{Q}{\epsilon_0 \epsilon_{22} S}$$

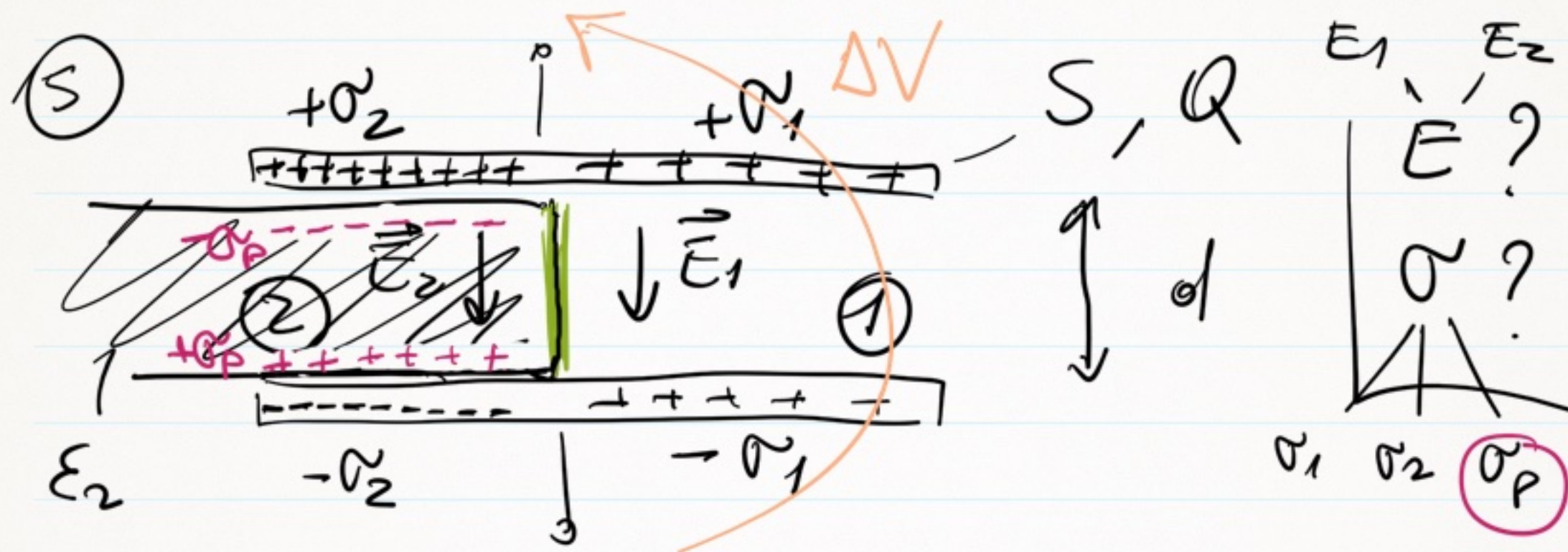
$$V = E_1 d_1 + E_2 d_2 = \frac{Q}{\epsilon_0 S} \left(\frac{d_1}{\epsilon_{21}} + \frac{d_2}{\epsilon_{22}} \right)$$

$$C = \frac{Q}{V} = \frac{\epsilon_0 S}{\frac{d_1}{\epsilon_{r1}} + \frac{d_2}{\epsilon_{r2}}} = \frac{\epsilon_0 \epsilon_{r1} \epsilon_{r2} S}{d_1 \epsilon_{r2} + d_2 \epsilon_{r1}}$$

$$(b) \quad \sigma_{p1} = \rho_1 = \epsilon_0 (\epsilon_{r1} - 1) E_1 = \cancel{\epsilon_0} (\epsilon_{r1} - 1) \frac{Q}{\cancel{\epsilon_0} \epsilon_{r1} S}$$

$$\sigma_{p2} = \rho_2 = \frac{\epsilon_{r2} - 1}{\epsilon_{r2}} Q$$

$$\sigma_p = \sigma_{p1} - \sigma_{p2} = \left(\frac{\epsilon_{r1} - 1}{\epsilon_{r1}} - \frac{\epsilon_{r2} - 1}{\epsilon_{r2}} \right) \frac{\epsilon_0 \epsilon_{r1} \epsilon_{r2} Q}{d_1 \epsilon_{r2} + d_2 \epsilon_{r1}}$$



$\rightarrow E_1 = E_2$ (cond. al brezzo)

$\rightarrow \Delta V_1 = \Delta V_2 \Rightarrow E_1 \neq E_2$ (equipotenziale)

$$\begin{cases} E_1 = \sigma_1 / \epsilon_0 & (\text{per Coulomb}) \\ E_2 = \sigma_{\text{tot}} / \epsilon_p = (\sigma_2 - \sigma_p) / \epsilon_0 \end{cases}$$

↳ tenendo conto del fatto che Gauss è soddisfacuto per l'insieme delle cariche: $\vec{\nabla} \cdot \vec{E} = \rho_{\text{tot}} / \epsilon_0$

$$\left\{ \begin{array}{l} \underline{\sigma_1} = \epsilon_0 E_1 = \epsilon_0 \cdot \underline{\frac{\Delta V}{d}} \end{array} \right.$$

$$\left\{ \begin{array}{l} \sigma_2 - \sigma_p = \epsilon_0 E_2 \end{array} \right., \text{ con } \underline{\sigma_p} = \epsilon_0 (\epsilon_2 - 1) E_2 = \epsilon_0 (\epsilon_2 - 1) \underline{\frac{\Delta V}{d}}$$

$$\rightarrow \underline{\sigma_2} = \epsilon_0 \frac{\Delta V}{d} + \epsilon_0 (\epsilon_2 - 1) \frac{\Delta V}{d} = \epsilon_0 \epsilon_2 \frac{\Delta V}{d} = \underline{\underline{\epsilon_2 \sigma_1}}$$

$$\Rightarrow \sigma_2 > \sigma_1$$

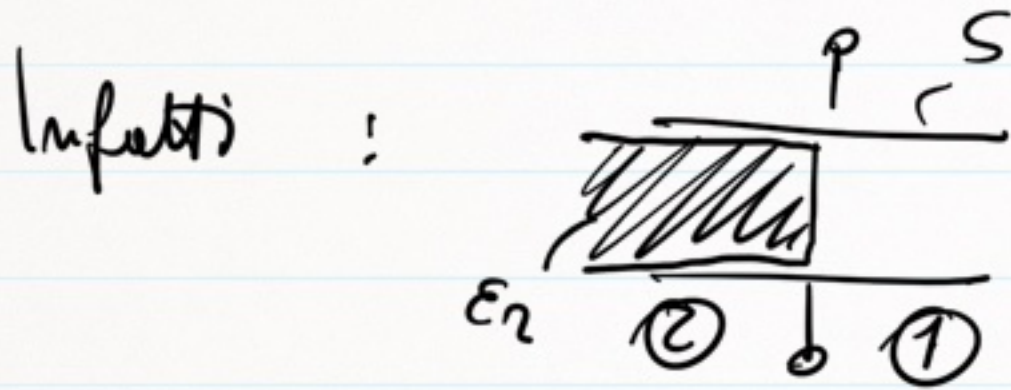
come ci aspettavamo!
per compensare la schermatura
permeabile dielettrica, e garantire
quindi la stessa ΔV (2) = (1)

$$\underline{\text{N.B.: } \sigma_p < \sigma_2}$$

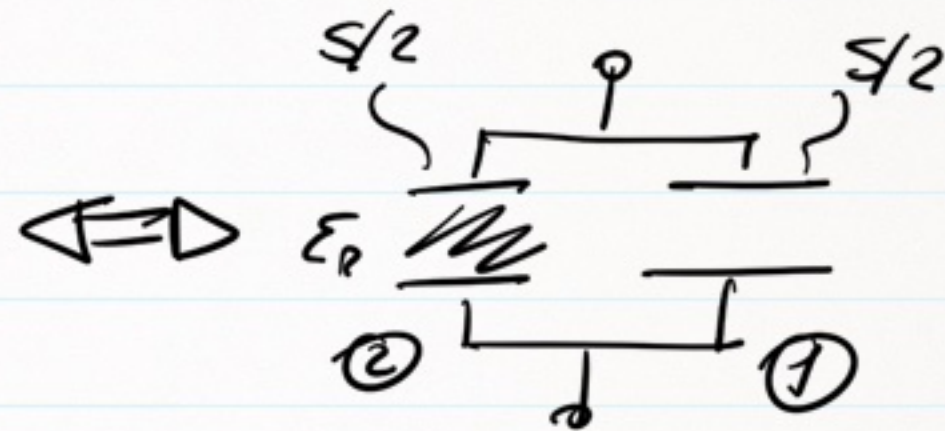
$$\vec{E}_0 \downarrow \uparrow \vec{E}_{\text{died}} = \downarrow \vec{E}_2$$

$$Q = \sigma_1 \cdot \frac{S}{2} + \sigma_2 \cdot \frac{S}{2} = \left(\frac{\epsilon_0 S}{2d} + \frac{\epsilon_0 \epsilon_2 S}{2d} \right) \Delta V$$

$$C = \frac{Q}{\Delta V} = \frac{\epsilon_0 S}{2d} + \frac{\epsilon_0 \epsilon_2 S}{2d}$$



C



$$C_{eq} = C_1 + C_2$$

$$C_1 = \frac{\epsilon_0 S}{2d}$$

$$C_2 = \frac{\epsilon_0 \epsilon_2 S}{2d}$$